# FIITJEE

# ALL INDIA TEST SERIES

## FULL TEST – XI

## JEE (Advanced)-2021

### PAPER –2

## TEST DATE: 26-09-2021

**Time Allotted: 3 Hours** **Maximum Marks: 198**

**General Instructions:**

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

***Section-A (01 – 06, 19 – 24, 37– 42)*** *this section contains* ***18 multiple*** *choice questions.*
*Each question has FOUR options.* ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
*For each question, choose the option(s) corresponding to (all) the correct answer(s)*
*Answer to each question will be evaluated according to the following marking scheme:*

***Full Marks*** **: +4** If only (all) the correct option(s) is (are) chosen:
***Partial Marks*** **: +3** If all the four options are correct but ONLY three options are chosen;
***Partial Marks*** **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
***Partial Marks*** **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
***Zero Marks*** **: 0** If none of the options is chosen (i. e. the question is unanswered);
***Negative Marks*** **: –2** In all other cases

***Section-B (07 – 12, 25 – 30, 43– 48)*** *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

***Section-C (13 – 18, 31 – 36, 49– 54)*** *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

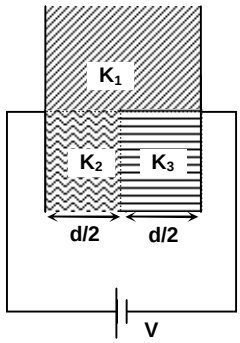
# Physics

## PART – I

### SECTION – A
### (One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A parallel plate capacitor is filled with three dielectric slabs of dielectric constant $K_1 = 2$, $K_2 = 3$ and $K_3 = 6$ as shown in the figure. The dielectric slab of dielectric constant $K_1$ covers the upper half area of the plates. A potential difference V is applied across the capacitor. (Given area of each plate of the capacitor = A, distance between the plates = d)



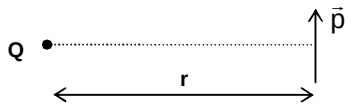

(A) If $Q_1$ and $Q_2$ be the magnitudes of charges on the upper half and lower half of one of the plates of the capacitor then $\frac{Q_1}{Q_2} = \frac{1}{2}$

(B) If $E_1$, $E_2$ and $E_3$ be the magnitudes of electric fields in dielectric of dielectric constant $K_1$, $K_2$ and $K_3$ respectively then $E_1 : E_2 : E_3 = 3 : 4 : 6$

(C) Ratio of energy densities in dielectrics $K_1$, $K_2$ and $K_3$ are 3 : 8 : 4 respectively.

(D) Capacitance of the capacitor is $C = \frac{3\varepsilon_0 A}{d}$

2. An infinitesimal electric dipole of dipole moment $\vec{p}$ is placed at a distance r from a point positive charge Q such that the direction of dipole moment $\vec{p}$ is perpendicular to the line joining the point charge Q and centre of the dipole. If the magnitudes of force and torque exerted by the point charge Q on the dipole are F and $\tau$ respectively, then



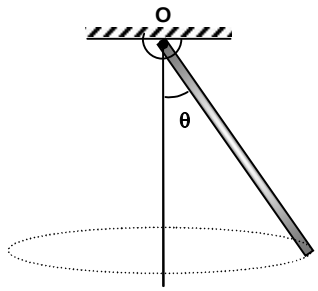

(A) $F = \frac{Qp}{4\pi\varepsilon_0 r^3}$

(B) $F = \frac{Qp}{2\pi\varepsilon_0 r^3}$

(C) $\tau = \frac{Qp}{2\pi\varepsilon_0 r^2}$

(D) $\tau = \frac{Qp}{4\pi\varepsilon_0 r^2}$

3. A thin uniform rod of mass m and length $\ell$ rotates with a constant angular velocity $\omega$ about the vertical axis passing through the rod's suspension point O. In doing so, the rod describes a conical surface with a half aperture angle $\theta$ as shown in the figure. Then choose the correct option(s).



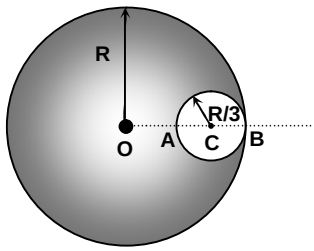
(A) Torque about O by centrifugal force acting on the rod (in the reference frame fixed to the rod) is $\frac{m\omega^2\ell^2\sin 2\theta}{3}$

(B) $\omega = \sqrt{\frac{3g}{2\ell\cos\theta}}$

(C) Component of reaction force at O in horizontal direction is $\frac{m\omega^2\ell\sin\theta}{2}$

(D) Magnitude of angular momentum of the rod about the axis of rotation is $\frac{m\omega\ell^2\sin^2\theta}{2}$

4. Consider a uniform spherical charge distribution of radius R and charge density $\rho$, centred at the origin O. In this charge distribution a spherical cavity of radius $\frac{R}{3}$ centred at C is made. The points A and B are on the surface of cavity lying on the straight line joining 'O' and 'C'. Let $V_A$, $V_B$ and $V_C$ are the electric potential at points A, B and C respectively, and $E_0$ and $E_C$ are the magnitude of electric field at O and C respectively. Then choose the correct option(s).




(A) $V_A - V_C = V_C - V_B$

(B) $(V_A - V_C) > (V_C - V_B)$

(C) $E_C = \frac{\rho R}{9\varepsilon_0}$

(D) $E_0 = \frac{\rho R}{36\varepsilon_0}$

5. An observer and a source of sound both are moving towards a reflector with the velocities of 38 m/s and 10 m/s respectively. The source is between the observer and the reflector. The reflector also moves away from the source with a velocity 15 m/s. The frequency of the sound emitted by the source is 340 Hz and the velocity of sound in air is 330 m/s. Then choose the correct option(s).

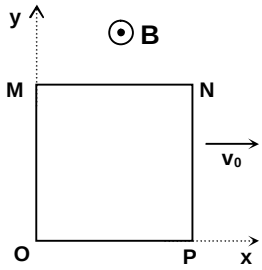
(A) Wavelength of echo (sound reflected from the reflector) received by the observer is approximately 1.03 m

(B) The beats frequency heard by the observer is 8 Hz.

(C) Wave length of sound heard by the observer coming directly from the source is 1 m.

(D) The frequency of the echo heard by the observer is 357 Hz

6. There exists a steady magnetic field $\vec{B} = B_0\left(1+\frac{x}{a}\right)\hat{k}$ throughout the space. A square conducting loop of side length a, mass m and resistance R is given a velocity $v_0$ along the positive x-axis at time t = 0 as shown in the figure. Then choose the correct option(s).




(A) Current in the loop as a function of velocity v of the loop, $I = \frac{B_0 a v}{R}$

(B) Current in the loop as a function of velocity v of the loop, $I = \frac{B_0 a v}{2R}$

(C) Velocity of the loop as a function of time is $v = v_0 e^{-\frac{B_0^2 a^2}{2mR}t}$

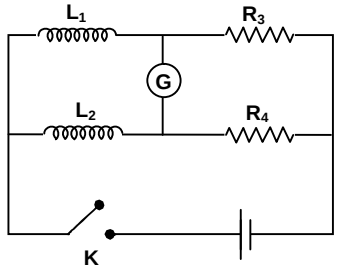
(D) Velocity of the loop as a function of time is $v = v_0 e^{-\frac{B_0^2 a^2}{mR}t}$

## SECTION – B
### (Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

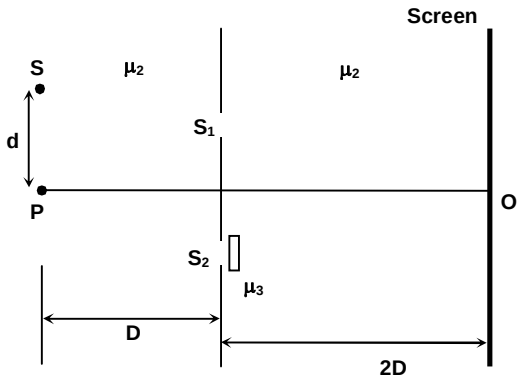
7. Two inductors of self inductances $L_1$ and $L_2$ and of resistances $R_1$ and $R_2$ (not shown here) respectively, are connected in the circuit as shown in the figure. At the instant t = 0, key K is closed. If the galvanometer shows zero deflection at all times after the key K is closed, then find the value of ratio $\left(\frac{L_1}{L_2}\right)$.

(Given $\frac{R_3}{R_4} = 2$)

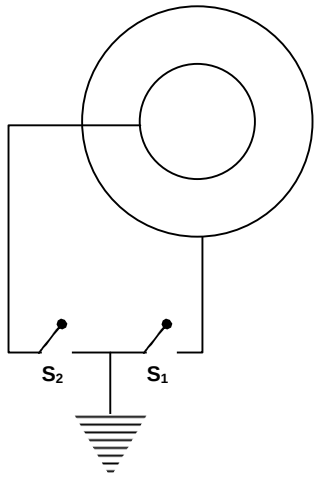



8. Electric field in a certain region is given as $\vec{E} = 4x\hat{i} - 3(y^2+2)\hat{j}$ N/C. Consider a cubical closed surface of side 1 meter such that its centre is at the origin and sides are parallel to the x, y and z axes respectively. If the net charge (in coulomb) enclosed by the cubical surface is $n\varepsilon_0$, find the value of n. (where $\varepsilon_0$ is permittivity of the free space)

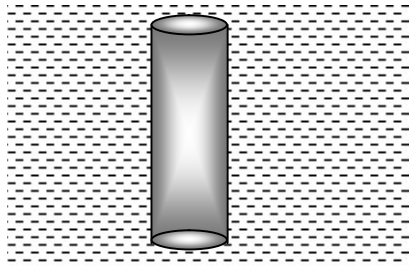
9. Consider the Young's double slit apparatus as shown in figure. The slits $S_1$ and $S_2$ are placed symmetrically about the line OP which is perpendicular to the screen. The entire space is filled with a liquid of refractive index $\mu_2$. A thin transparent sheet of thickness t and refractive index $\mu_3$ is placed in front of the slit $S_2$. A monochromatic source S is placed at a distance d above the line OP. The separation between the slits $S_1$ and $S_2$ is d. Given D = 1m, d = 2mm, t = 6μm, $\mu_2$ = 1.2 and $\mu_3$ = 1.8. Find the distance of central maximum (in mm) from the origin O.



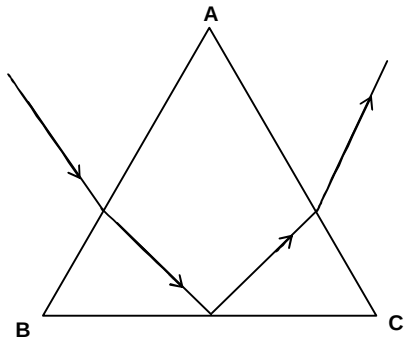

10. Two concentric conducting shells of radii R and 2R are shown in the figure. Initially a charge q is imparted to the inner shell. Let the potential difference between the cells is 10V. The switches $S_1$ and $S_2$ are alternatively closed 3 times each. After this if the magnitude of potential difference between the shells is $\frac{n}{4}$ volts, find the value of n. (Note that finally the switch $S_2$ is closed)



11. A uniform solid cylinder made of material of density 300 kg/m$^3$ is kept at rest inside the water vertically so that its top surface touches the free surface of the water. The length of the cylinder is 67.50 cm. Now the cylinder is released. Find the velocity (in m/s) of the cylinder when it leaves the water surface (Given density of water = 1000 kg/m$^3$, g = 10 m/s$^2$)



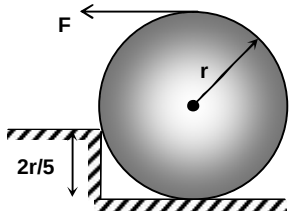
12. The path of a ray of light passing through an equilateral glass prism ABC is shown in the figure. The ray of light is incident on the face BC at just greater than critical angle for total internal reflection. The total angle of deviation after the refraction at the face AC is 108°. The refractive index of the material of the prism is $\mu$. If $\sqrt{\mu^2-1}=\frac{n}{5\sqrt{3}}$. Then find the value of n. (Given sin24° = 0.4)



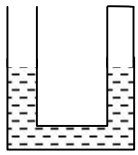

## SECTION – C
## (Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A solid cylinder of mass m and radius r is placed at the bottom of a fixed step of height $\frac{2r}{5}$ as shown in the figure. A horizontal force F is applied on the topmost point of cylinder so that it just starts climbing the step without slipping. Find the minimum coefficient of static friction between the cylinder and step so that the cylinder does not slip just after the application of force F.

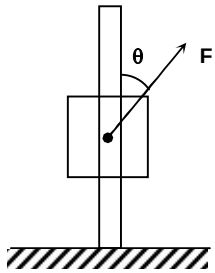



14. A thin U-tube sealed at one end consists of three bends of length $\ell$ = 0.50 m each, forming right angles. The vertical parts of the tube are filled with mercury to half the height. All the mercury is displaced from tube by heating slowly the gas in the sealed end of the tube, which is separated from the atmospheric air by mercury. Find the work done (in joule) by the gas. Given atmospheric pressure $P_0 = 10^5$ Pa, density of mercury = $13.6 \times 10^3$ kg/m$^3$ and cross sectional area of the tube is A = 1 cm$^2$.



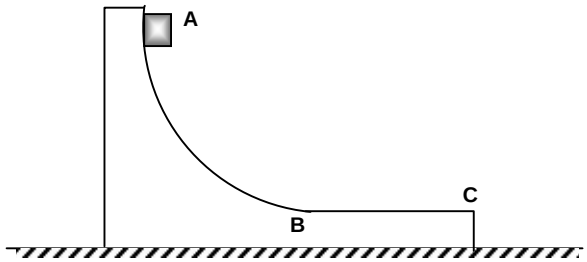
15. Consider a frictionless tunnel is dug along a diameter of the earth. An object is projected from the surface of the earth in the tunnel towards the centre of the earth with a speed $v_0=\sqrt{gR}$, where R is radius of the earth and g is the acceleration due to gravity at the surface of the earth. If the time taken by the object to reach the centre of the earth is $n\pi\sqrt{\frac{R}{g}}$, then find the value of n.

16. The collar of mass m slides up a vertical shaft under the action of a force F of constant magnitude but variable direction. If the angle made by the force F with the vertical at any time t is $\theta = kt$ where k is a positive constant and collar starts from $\theta = 0°$. The collar comes to rest as $\theta$ reaches the value $\pi/2$. The coefficient of kinetic friction between the collar and shaft is $\mu_k = 1/3$. If the value of $F = n\pi mg$, then find the value of n.




17. The half lives of radio isotopes $P^{32}$ and $P^{33}$ are 15 days and 25 days respectively. These radioisotopes are mixed in the ratio of 3 : 2 of their atoms. If the initial activity of the mixed sample is 4.48 mCi, then find the activity (in mCi) of the mixed isotopes after 75 days.

18. A wedge of mass 2m is placed on a smooth horizontal surface. The frictionless surface AB of the wedge is in the shape of quarter of circle of radius r = 0.72 m. The horizontal surface BC is rough and long. A block of mass m is released from rest at the top of the wedge. It slides down and comes to rest relative to the wedge after covering a horizontal distance $\ell$ on the rough horizontal surface of the wedge. The coefficient of friction between the block and the horizontal surface of the wedge is 0.5. Find the value of $\ell$ (in meter).



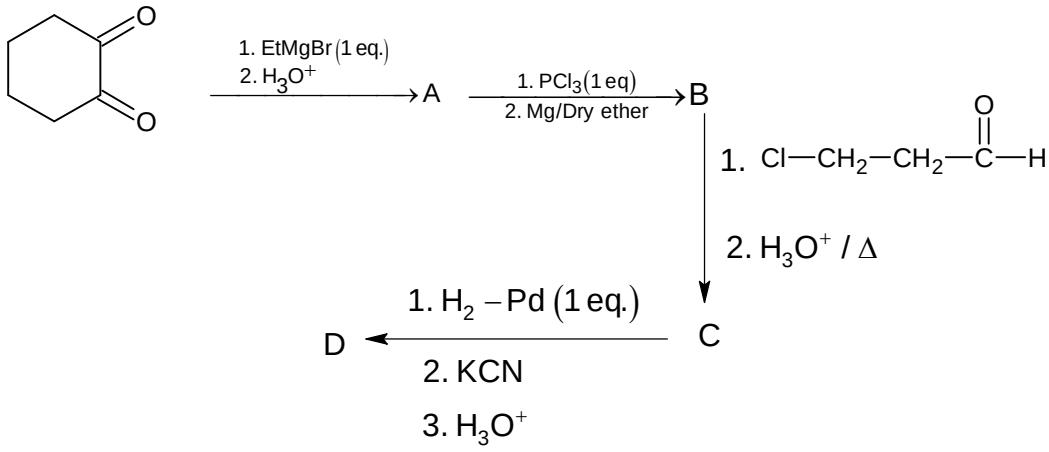
# Chemistry

## PART – II

### SECTION – A
### (One or More than one correct type)

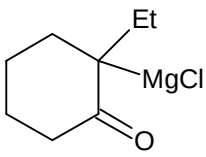
This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19.



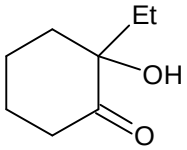

(A)

A is:



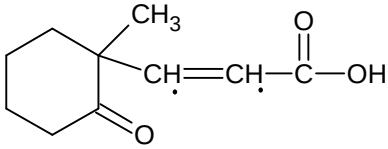

(B)

A is:



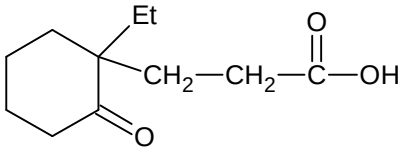

(C)

B is:



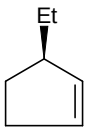

(D)

D is:




20.

(P)  $\xrightarrow[CCl_4]{Br_2}$ The products are

(Q)  $\xrightarrow[CCl_4]{Br_2}$ The products are

(A) P gives meso compound, Q gives diastereomers
(B) Both 'P' and 'Q' gives enantiomers
(C) 'P' gives enantiomers where as Q gives diastereomers
(D) Product of 'P' and 'Q' are optically active

21. Which of the following is/are intensive property?
(A) $E_{cell}$
(B) $E^{o}_{cell}$
(C) Normality
(D) Entropy

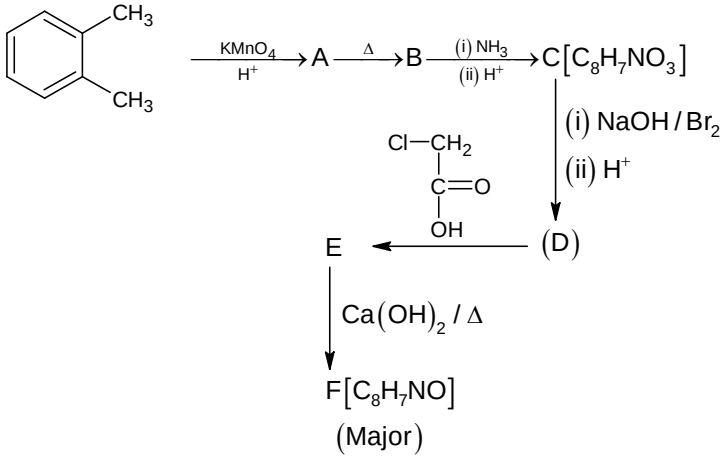
22. Which of the following is/are low spin complex?
(A) $\left[Co(H_2O)_6\right]^{3+}$
(B) $\left[Co(H_2O)_6\right]^{2+}$
(C) $\left[Fe(NH_3)_6\right]^{3+}$
(D) $\left[Co(H_2O)_3F_3\right]$

23. $P_4 + SOCl_2 \longrightarrow PCl_3 + SO_2 + A$
$A + Cl_2 + NH_3 \longrightarrow B + HCl$
$A + NH_4Cl \longrightarrow B$
(A) A is $SCl_4$
(B) A is $S_2Cl_2$
(C) B is $S_4N_4$
(D) B is when heated at $130^{o}C$ $N_2$ will obtained

24. Which of the following is/are correct?
(A) $XeF_6 + SiO_2 \longrightarrow XeOF_4 + SiF_4$
(B) $XeOF_4 + H_2 \xrightarrow{300^{o}C} Xe + H_2O + HF$
(C) $XeO_2F_2 + H_2 \xrightarrow{300^{o}C} Xe + H_2O + HF$
(D) $XeO_3$ in strongly basic medium disproportionate to form perxenate ion and elemental Xe

## SECTION – B
## (Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

25. 

Unsaturation factor in major product is

26. Minimum number of hydrogen atom to get maximum number of different spectral lines for the transition n = 4 to n = 1 is

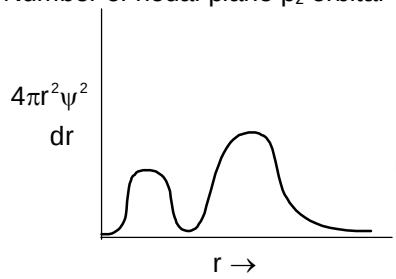
27. Sum of peroxide linkage present in $K_3CrO_8$ and $CrO_5$ is

28. Total number of correct statements among the following is

(i) $(Mn)_x (Co)_y$ the value of $\frac{x+y}{2}$ according to 18 electron rule is 6.

(ii) On strong heating of $H_3BO_3$ gives $B_2O_3$.

(iii) Number of nodal plane $p_z$ orbital = 1

(iv) 

is for 3p-orbital.

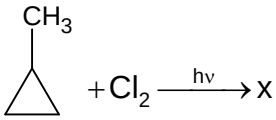
29. Number of black colour sulphide among the following is
$CdS, SnS, NiS, CoS, MnS, PbS, SnS_2, As_2S_3, CuS$

30. Number of ortho oxo acid from following is
$H_2CO_3, H_2SO_4, H_2SO_3, H_3PO_4, H_3BO_3, H_3PO_3, HNO_3, HNO_2$

## SECTION – C
## (Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

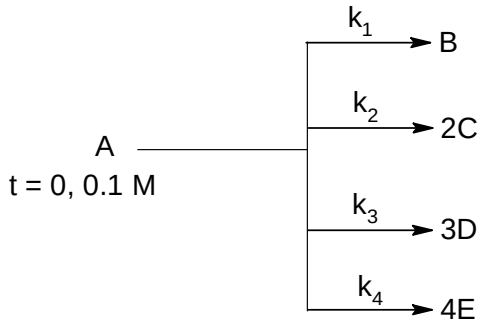
31. A triatomic linear ideal gas undergoes a process in which the ratio of P to $V^2$ at instant is constant and equal to 1. The molar heat capacity of the gas is [R = 8.4 J/K mol) in J/K mol.

32. 

(Whereas total number of diahalogen derivatives excluding optically inactive dihalogen derivatives). Then value of $\frac{x}{10}$ is ?

33. An aqueous solution of metal bromide $MBr_2\,(0.01\,M)$ is saturated with $H_2S$, the minimum $\left[H^+\right]$ at which MS will precipitate?
$K_{sp}$ for $MS = 4.0\times10^{-21}$, concentration of saturated $H_2S = 0.1\,M$, $k_1 = 10^{-7}, k_2 = 1.6\times10^{-12}$ for $H_2S$.

34.




Given : $k_1 : k_2 : k_3 : k_4 = 1 : 2 : 3 : 4$

What is the value [E] after time $t = \infty$. [Above reaction follow the 1st order kinetics]

35. The mass of ice that will be separated out on cooling a solution containing 40 g of ethylene glycol in 400 gram of water to $-5^{\circ}C$ is. $\left(K_f \text{ for water} = 1.86\ K\,mol^{-1}\ kg\right)$

36. The reduction electrode potential of (in volt)

$E_{S^{2-}(aq)|Ag_2S|Ag}$ in solution buffered at pH = 4 and which is also saturated with

$0.01\,M\ H_2S\left[K_{a_1} = 10^{-7}, K_{a_2} = 10^{-13}\right]$ for $\left[K_{sp}\right]_{Ag_2S} = 4 \times 10^{-44}$, $E^{o}_{Ag^{+}/Ag} = 0.9$ volt

$\log 2 = 0.3,\ 2.303\dfrac{RT}{F} = 0.06$

# Mathematics

## PART – III

### SECTION – A
### (One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let f(x) be a function f : R $\to$ R, such that $f(x)=\begin{cases}\left(\dfrac{a_1^x+a_2^x+a_3^x+.....+a_n^x}{n}\right)^{\frac{1}{x}} & , \ x\neq 0\\ (a_1a_2a_3\ .....\ a_n)^{\frac{1}{n}} & , \ x=0\end{cases}$ where $a_1$, $a_2$, $a_3$, ....., $a_n > 0$, then which of the following is/are **CORRECT?**

(A) $\lim\limits_{x\to 0} f(x) = f(0)$

(B) $\lim\limits_{x\to 0} f(x) = f(1)$

(C) $f(2) \geq f(-1)$

(D) $f(1) \geq f(0)$

38. Let $z_1$ and $z_2$ be two complex numbers satisfying the equation $12|z|^2 = 2|z + 2|^2 + |z^2 + 1|^2 + 31$, then

(A) $|z_1 - z_2| = 4\sqrt{5}$

(B) $|z_1 + z_2| = 6$

(C) $|z_1 - z_2| = 2\sqrt{5}$

(D) $|z_1 + z_2| = 2$

39. Let $S_n$ be the sum of the first n terms of the arithmetic sequence $b_1$, $b_2$, ..... $b_n$, such that $f(i, j, k)=\dfrac{S_i}{i}(j-k)+\dfrac{S_j}{j}(k-i)+\dfrac{S_k}{k}(i-j)\ \ \forall\ i, j, k \in N$, then

(A) f(1, 3, 5) = 15

(B) f(2, 7, 12) < 21

(C) f(1, 3, 5) = 21

(D) f(2, 7, 12) < 15

40. If position vectors of the five points A, B, C, D and M are $\hat{i}+\hat{j}$, $\hat{i}+3\hat{j}$, $2\hat{i}-\hat{j}$, $-3\hat{i}-3\hat{j}$ and $\hat{j}+\hat{k}$ respectively and P, Q, R and S are midpoints of the line segments AB, BC, CD and DA respectively, then

(A) volume of tetrahedron MPSA is, $\frac{1}{8}$ unit$^3$

(B) volume of tetrahedron MRSD = $\frac{1}{3}$ volume of tetrahedron MACD

(C) volume of tetrahedron MRQC = $\frac{1}{4}$ volume of tetrahedron MCBD

(D) area of quadrilateral ABCD = 2 area of quadrilateral PQRS

41. If $P = \lim\limits_{n\to\infty} \frac{1}{n^4} \sum\limits_{1\le i<j\le n} ij$, $Q = \lim\limits_{n\to\infty} \frac{1}{n^6} \sum\limits_{1\le i<j\le n} i^2j^2$ and $R = \lim\limits_{n\to\infty} \frac{1}{n^2} \sum\limits_{1\le i<j\le n} \cos\left(\frac{i}{n}\right)\cos\left(\frac{j}{n}\right)$, then which of the following is/are **CORRECT?**

(A) $P + Q < \frac{1}{2}$

(B) $P - Q < \frac{1}{12}$

(C) $P + Q < \frac{5}{8}$

(D) $Q - R < \frac{4}{9}$

42. If $A_n = \int_0^{\infty} \left[\frac{n}{e^x}\right] dx$, $n \in N$ (where [.] denote greatest integer function), then

(A) $\lim\limits_{m\to\infty} \frac{1}{m} A_m = 0,\ m \in N$

(B) $A_3 = 2\ell n\frac{3}{2}$

(C) $A_2 = \ell n 2$

(D) $\lim\limits_{m\to\infty} \frac{1}{m} A_m = 1,\ m \in N$

## SECTION – B
## (Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

43. A real sequence is define as $t_1 = 20$, $t_2 = 21$ and $t_n = \frac{5t_{n-1} + 1}{25t_{n-2}}$ for all $n \ge 3$ and if $5t_{13} = \frac{p}{q}$ (where p and q are coprime), then $|p - q|$ is

44. The number of integral values that belongs to the exhaustive set of domain of the function $f(x) = \sqrt{(2x+9) - \left(\frac{2x}{1-\sqrt{1+2x}}\right)^2}$ is

45. There are 64 cells (each cells of dimension $1 \times 1$ unit square) on $8 \times 8$ chess board, if number of ways of selecting a pair of cells from 64 cells so that they share atleast one common vertex is P, then number of positive divisor of $\frac{P}{2}$ is

46. Let f(x) be a function, $f : R \to R$ such that $f(x)f(y) + f(xy) \le x + y\ \forall\ x, y \in R$, then $(f(1))^2 + (f(-1))^2$ is equal to

47. Let x, y, z be three real numbers such that $x^2 + y = z^2$, $y^2 + z = x^2$, $z^2 + x = y^2$, then the value of $x^3 + y^3 + z^3$?

48. If p and q positive integers and $\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\dfrac{\sin^2\left(\dfrac{10^\circ}{3^r}\right)}{\cos\left(\dfrac{30^\circ}{3^r}\right)}=\dfrac{1}{p+\sqrt{q}}$, then the value of $\left[\sqrt{q}-p\right]$ is (where [.] denotes greatest integer function) is

## SECTION – C
## (Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. The number of integer values of k, such that equation $\left|\dfrac{k-|20|x|-x^2|}{21}\right|=1$ has 12 distinct real solutions

50. Let $a_1, a_2, \ldots, a_6$ are six non negative real numbers such that $\sum\limits_{i=1}^{6}a_i=1$ and $a_1a_3a_5 + a_2a_4a_6 \geq \dfrac{1}{540}$, let the maximum value of $a_1a_2(a_3 + a_6) + a_2a_3(a_4 + a_5) + a_5a_6(a_1 + a_2)$ is $\dfrac{m}{n}$, m, n are coprime, then (n – m) is equal to

51. A triangle whose coordinate of vertices are $A(a_1, b_1)$, $B(a_2, b_2)$, $C(a_3, b_3)$ and centroid is (1, 1) given that lengths AB : BC : CA = 3 : 4 : 5 and $a_1^3+a_2^3+a_3^3-3a_1a_2a_3=20$, $b_1^3+b_2^3+b_3^3-3b_1b_2b_3=21$. If area of triangle ABC is $\dfrac{p}{q}$, p, q are coprime, then p – q is equal to

52. If $(x_1, y_1)$ and $(x_2, y_2)$ are two distinct points satisfying the curve $(x – 3)(x^2 + y^2) + 3x – y = 0$ and $y(x^2 + y^2) – (x + 3y) = 0$, then the value of $\sum\limits_{i=1}^{2}x_i^4+y_i^4$ is

53. In $\Delta ABC$, AC = 3AB. The internal angle bisector of $\angle BAC$ meets BC at D. The circle with AC as diameter cuts the extension of AD at E, then $\dfrac{AD}{AE}$ is

54. If n is the number of integral values of P, $P \in R$, such that the equation $\sqrt{t^2-P}+\sqrt{4t^2-4}=t$ has a solution, then 50n is equal to